

NUMERICAL ANALYSIS

Numerical integration: the trapezoidal rule

A short mathematical note

1 From a line segment to a quadrature rule

Suppose we want to approximate

$$I(f) = \int_a^b f(x) \, dx.$$

Replace the graph of f by the straight line joining its endpoint values. Integrating this linear interpolant gives the trapezoidal approximation

$$T_1(f) = \frac{b-a}{2} (f(a) + f(b)). \quad (1)$$

The rule is exact for affine functions. A curved graph introduces an error that can be controlled by the second derivative.

2 The composite rule

Divide the interval into n equal subintervals, with

$$h = \frac{b-a}{n}, \quad x_j = a + jh, \quad j = 0, \dots, n.$$

Apply (1) on each subinterval and add the contributions:

$$\begin{aligned} T_n(f) &= \sum_{j=0}^{n-1} \frac{h}{2} (f(x_j) + f(x_{j+1})) \\ &= h \left[\frac{f(a)}{2} + \sum_{j=1}^{n-1} f(x_j) + \frac{f(b)}{2} \right]. \end{aligned} \quad (2)$$

Interior values occur twice, so their weights become h . Each endpoint occurs only once and receives weight $h/2$.

3 An error estimate

Theorem 1. *If $f \in C^2([a, b])$, there is a point $\xi \in (a, b)$ such that*

$$I(f) - T_n(f) = -\frac{(b-a)h^2}{12} f''(\xi). \quad (3)$$

Consequently,

$$|I(f) - T_n(f)| \leq \frac{(b-a)h^2}{12} \max_{x \in [a, b]} |f''(x)|.$$

Proof. On $[x_j, x_{j+1}]$, the linear interpolation remainder is

$$f(x) - p_j(x) = \frac{f''(\eta_x)}{2} (x - x_j)(x - x_{j+1}).$$

The product has one sign on this interval. The weighted mean-value theorem for integrals therefore gives a local error $-h^3 f''(\xi_j)/12$. Summing over all intervals and using continuity of f'' yields (3). $\square$

4 A worked example

Take $f(x) = x^2$ on $[0, 1]$. The exact integral is $I(f) = 1/3$. Since $f'' = 2$, the error formula gives

$$T_n(f) = \frac{1}{3} + \frac{1}{6n^2}. \quad (4)$$

The approximation is too large, as expected for a convex graph: each chord lies above the curve.

n	$T_n(f)$	$T_n(f) - I(f)$
1	0.50000000	0.16666667
2	0.37500000	0.04166667
4	0.34375000	0.01041667
8	0.33593750	0.00260417
16	0.33398438	0.00065104

Doubling the number of subintervals divides the error by four. This is the characteristic behavior of a method with second-order accuracy.

5 Richardson extrapolation

For a sufficiently smooth integrand, suppose the leading error has the expansion

$$T_n(f) = I(f) + Ch^2 + O(h^4).$$

Refining the mesh gives

$$T_{2n}(f) = I(f) + \frac{C}{4}h^2 + O(h^4).$$

The combination

$$R_n(f) = \frac{4T_{2n}(f) - T_n(f)}{3} = I(f) + O(h^4) \quad (5)$$

eliminates the leading error term. For the quadratic example, inserting (4) into (5) gives $R_n(f) = 1/3$ exactly.

6 Practical reading of the estimate

- The regularity assumption matters. A nonsmooth integrand may not exhibit the predicted convergence rate.
- A small difference between two approximations is evidence of convergence, not by itself a rigorous error bound.
- Formula (3) describes discretization error. Floating-point roundoff is a separate source of error.

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